

LOTFI AIT BAAYA

DATA ANALYST & AI / ML DEVELOPER



CONTACT

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SKILLS

Data Analysis

- Python, Pandas, NumPy, SQL

Programming

- Python, JavaScript, C/C++

Data Visualization

- Matplotlib, Seaborn

Web & Development

- React.js, Git, GitHub

Machine Learning & AI

- Scikit-learn, PyTorch, OpenCV, MediaPipe

EDUCATION

Bachelor's Degree — Data Analysis & Artificial Intelligence

Faculty of Sciences, Agadir, Morocco
Sept 2024 – Mar 2026

Associate Degree — Computer Engineering

Polydisciplinary Faculty, Taroudant, Morocco
Sept 2022 – Jul 2024

CERTIFICATIONS

Data Science Essentials with Python

Cisco — August 2026

LANGUAGES

Arabic	Native
French	Fluent
English	Fluent

PROFILE

Data Analyst and AI/ML Developer with a Bachelor's degree in Data Analysis & Artificial Intelligence and an Associate degree in Computer Engineering. Built 16+ data analysis projects spanning business, environmental, health, transportation, and technology domains, and developed AI applications including a real-time hand gesture recognition system and a browser-based self-driving car simulation with a neural network built from scratch. Hands-on experience in machine learning, computer vision, data analysis, and statistical analysis, with an interest in applying AI and digital technologies to support data-driven decision-making and digital transformation in development contexts.

PROJECTS

Self-Driving Car Simulation

React.js · HTML5 Canvas · JavaScript – Neural Network from Scratch

Browser-based simulation of a 3-lane highway environment, built to explore neural network fundamentals without relying on ML libraries. Implemented a feedforward network ($6 \rightarrow 12 \rightarrow 3$, ReLU + softmax) with forward pass, backpropagation, and gradient descent written entirely in vanilla JavaScript, trained on a manually labeled set of 24 lane-occupancy scenarios. Added procedural traffic generation, object pooling, real-time collision detection, and JSON-based model persistence.

Hand Gesture Classifier & Mouse Control

Python · MediaPipe · OpenCV · Scikit-learn · PyAutoGUI

Real-time computer vision pipeline that tracks 21 hand landmarks to classify four gestures and translate them into cursor control. Compared Logistic Regression and Random Forest classifiers with cross-validation, applied feature normalization, and used an exponential moving average to smooth cursor movement.

Data Analysis Portfolio — 16+ Projects

Python · Pandas · Scikit-learn · Statistical Analysis

Completed 16+ analytical projects involving data cleaning, exploratory data analysis, statistical analysis, regression, hypothesis testing, visualization, and insight communication across business, environmental, health, transportation, and technology topics.

Selected analytical findings:

- Lion Attacks:** Observed a negative correlation between evening moonlight and reported attacks.
- E-Bike Stopping Distances:** Improved regression fit from R^2 0.968 to 0.989 using a quadratic model.
- Global Internet Adoption:** Analyzed worldwide adoption trends and regional differences, including the digital divide in Africa.
- Flight Delays:** Identified differences in delay rates across weekdays and examined the relationship between passenger volume and delays.
- Plant-Based Coffee Shop:** Analyzed 1,170 survey responses to identify consumer preferences.